CATEGORICALLY COMPACT OBJECTS OF QCOH ARE EXACTLY THE FINITELY PRESENTED MODULES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 0FWX: sources/stacks-chow/tags.txt, source lines 19862–19862. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01BS: sources/stacks-chow/tags.txt, source lines 1737–1737. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01PK: sources/stacks-chow/tags.txt, source lines 2233–2233. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then $\mathcal{F}$ is a categorically compact object of $QCoh(\mathcal{O}_X)$ if and only if $\mathcal{F}$ is of finite presentation.*

Proof. See Categories, Definition 0FWX for our notion of categorically compact objects in a category. If $\mathcal{F}$ is of finite presentation then it is categorically compact by Modules, Lemma 01BS. Conversely, any quasi-coherent module $\mathcal{F}$ can be written as a filtered colimit $\mathcal{F} = \operatorname{colim} \mathcal{F}_i$ of finitely presented (hence quasi-coherent) $\mathcal{O}_X$-modules, see Properties, Lemma 01PK. If $\mathcal{F}$ is categorically compact, then we find some i and a morphism $\mathcal{F} \to \mathcal{F}_i$ which is a right inverse to the given map $\mathcal{F}_i \to \mathcal{F}$. We conclude that $\mathcal{F}$ is a direct summand of a finitely presented module, and hence finitely presented itself. □

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